

RESEARCH PROPOSAL

PhysMon: Hidden-State Detection of Physics Shortcut Sensitivity under Solver-Verified Counterfactuals

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Abstract

Language models can produce fluent, step-by-step chain-of-thought solutions to physics problems while silently relying on physically irrelevant cues rather than governing principles. This proposal presents **PhysMon**, a research programme that will investigate whether prompt-side hidden states can predict model-specific shortcut sensitivity under solver-verified, physics-preserving counterfactual edits—*before* that sensitivity becomes visible in generated reasoning or final answers.

We will construct a family-structured benchmark of 120–180 canonical physics problem families in classical mechanics and electrostatics, where designated cue edits provably preserve the correct answer according to a symbolic or executable verifier. A core behavioural study will measure how much model output distributions shift across invariant families—a property we term *counterfactual shortcut sensitivity*—and will test whether prompt-side hidden states predict this sensitivity more reliably than surface classifiers, CoT-text monitors, confidence measures, and black-box perturbation baselines. A carefully controlled causal study will test whether localized prompt-side representations contribute to cue dependence. The primary targets are ACL/EMNLP/NAACL framed around counterfactual scientific-reasoning monitorability; ICLR is a stretch target contingent on unusually clean causal evidence. All analysis uses open-weight models with locally available GPU resources; no commercial API credits are required.

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1 Motivation and Problem Statement

Large language models now produce fluent chain-of-thought (CoT) solutions to undergraduate-level physics problems, achieving non-trivial scores on benchmarks such as UGPhysics (et al., 2025c), PhysReason (et al., 2025f), and PhysUniBench (et al., 2025b). Yet benchmark accuracy alone cannot distinguish two qualitatively different computational strategies: a model that applies the governing physical law *faithfully*, versus a model that exploits a physically irrelevant cue present in the prompt—a surface correlate, a redundant variable, or a distractor—and coincidentally reaches the correct answer.

This distinction is consequential. A model that reasons faithfully generalises when cue structures change; a model that exploits shortcuts fails *silently* when those cues are absent, altered, or misleading. Silent brittleness is especially dangerous in scientific and educational contexts: the model’s written explanation may appear physically sound even when the underlying computation does not reflect the stated principle.

Existing evaluation frameworks measure what the model outputs, not how that output was computed. Recent work establishes that CoT explanations are often incomplete or post-hoc rationalisations of the actual computation (Chen et al., 2026; et al., 2026a; Turpin et al., 2023). Confidence and self-consistency metrics are insensitive to *which* cue was used. External judge-based faithfulness scorers are noisy and can be misled by stylistically plausible reasoning text (et al., 2026b).

Why physics. Unlike generic NLP tasks, physics problems carry verifiable intermediate structure: dimensional homogeneity, conservation laws, coordinate independence, and algebraic identities. This structure enables construction of *controlled counterfactual families*: sets of semantically equivalent prompts where a designated physically irrelevant variable is edited while the solver-verified correct answer remains unchanged. Such families are rare in NLP. They allow the researcher to certify, independently of any model, that the correct answer *must* be invariant across the family—making “shortcut sensitivity” an objective, model-specific behavioural property rather than an annotation artefact.

The core scientific question. Can prompt-side hidden states predict model-specific shortcut sensitivity before any failure is visible in the model’s generated reasoning or final answer? If yes, hidden-state monitors could detect silent brittleness in scientific reasoning, quantify a faithfulness gap invisible in outputs, and—if causal evidence is sufficient—provide a representation-level account of cue dependence.

Project scope. This is an *independent secondary research project*, designed to run in parallel with a primary research programme and not to merge with or depend on unrelated agentic or multi-agent projects. A possible future application—using the monitor as a trigger for solver escalation or verifier-agent routing—is listed in Section 18 as optional future work only and does not form part of the core scope.

2 Research Gap and Novelty

2.1 What the Literature Already Establishes

Three claims are well-supported in 2025–2026 and *cannot* constitute novelty:

1. **CoT is often an incomplete window into model computation.** Multiple papers show that models can reach correct or incorrect answers while their written reasoning omits the actual driving factor or provides a post-hoc rationalisation (Chen et al., 2026; et al., 2026a,b; Turpin et al., 2023).
2. **Hidden states carry reasoning-relevant signals that surface text or confidence metrics miss.** Probes trained on intermediate activations can predict final-answer correctness, reasoning errors, and hidden task-relevant information before it appears in text (Allen-Zhu and Li, 2025; et al., 2025a, 2026d, 2024, 2025e; Mallen et al., 2023).
3. **Physics and science reasoning are already active benchmark areas.** Recent benchmarks document frontier-model limitations on university physics, making “LLMs are imperfect at physics” background knowledge rather than a contribution (et al., 2025b,c,f).

2.2 What Remains Open

Despite the above, a scientifically meaningful gap exists:

- No existing work constructs a **solver-verified, family-structured physics benchmark** in which designated cue edits provably preserve the correct answer and the benchmark is designed to measure *model-specific* counterfactual shortcut sensitivity.
- No existing work demonstrates that **prompt-side hidden states predict behaviourally defined counterfactual shortcut sensitivity**—as opposed to prompt condition, generic correctness, or output confidence—in a physics domain with rigorous artefact controls and family-level held-out evaluation.
- The **causal question**—whether localised prompt-side representations contribute to cue dependence under solver-verified invariant edits—has not been addressed with the combination of necessity, sufficiency, and randomised-direction controls required for a credible result.

2.3 Differentiation from Closest Adjacent Work

Table 1 summarises the most relevant adjacent work. Four specific threats require explicit attention:

- et al. (2025g) (SenseMath) is the closest dataset precedent in spirit—a controlled numerical shortcut benchmark—but does not target physics-specific invariances, uses no symbolic/solver verification, and includes no internal activation analysis.
- et al. (2024) and et al. (2025a) establish that hidden states predict reasoning errors and carry recoverable task information. They target generic correctness, not physics-

specific cue dependence; PhysMon must demonstrate that its monitor is not reducible to a generic difficulty or error-prediction signal.

et al. (2026e) demonstrates that near-perfect probe accuracy can collapse to chance after residualising format confounds, directly attacking prompt-condition probing. PhysMon addresses this by using *behaviourally derived labels*, matched controls, and family-level held-out evaluation.

et al. (2026c) shows that many purported SAE reasoning features collapse under lexical falsification tests. This motivates aggressive artefact falsification throughout and demotion of SAE-based analysis to an optional extension.

Table 1: Selected adjacent work, its relationship to PhysMon, and the gap that remains. Papers are grouped by contribution type.

Paper(s)	Relationship to PhysMon	Gap / required differentiation
Turpin et al. (2023) ; Chen et al. (2026) ; et al. (2026a)	Establish CoT unfaithfulness	Generic claim is established; PhysMon must show a physics-specific, objectively verifiable faithfulness gap
et al. (2026b)	Formalises CoT faithfulness as information flow	Provides a useful conceptual frame; PhysMon should align with this language
et al. (2024) ; et al. (2025a) ; et al. (2026d)	Hidden states predict reasoning quality and carry recoverable task information	Target generic correctness/error; PhysMon must show its sensitivity target is not reducible to general difficulty
Wang et al. (2023) ; Conmy et al. (2023)	End-to-end circuit analysis and automated patching	Raise the bar for mechanistic claims; PhysMon targets cue sensitivity, not a full reasoning circuit
et al. (2026e) ; et al. (2026c)	Probes may detect format; SAE features may be lexical artefacts	Direct threat; addressed by behaviourally derived labels and matched controls
et al. (2025g)	Controlled numerical shortcut benchmark	No physics-specific invariances; no internal analysis; no solver certification
Li et al. (2024) ; et al. (2025d)	Shortcut benchmarks in reasoning tasks	Physics adds verifier-backed constructs; internal analysis differentiates

3 Formal Problem Definition

3.1 Three Levels of Analysis

A central conceptual correction relative to earlier formulations of this project is the explicit distinction between three levels that must not be conflated:

1. **Prompt condition.** Which cue value or rendering z_i is present in the prompt. This is a property of the *stimulus*, not of the model.
2. **Observed model behaviour.** How much the model’s answer, answer distribution, or log-probability assigned to the solver-verified answer changes across a family of prompts whose correct answer should remain invariant. This is a model-specific, behaviourally observable property.
3. **Internal representation or mechanism.** A hidden-state signal that predicts the behavioural sensitivity (Level 2), or a representation whose controlled intervention changes that sensitivity. This is the object of mechanistic investigation.

The primary probe and monitor target in this project is **Level 2** (model-specific behavioural shortcut sensitivity), not Level 1 (cue presence). Level 3 interventions are subject to the highest evidential standards and are treated as a higher-risk secondary objective. The probe must predict *future counterfactual instability*, not merely decode which cue value is present in the current prompt.

3.2 Notation

Let τ denote a *canonical physics template* characterised by:

$$\tau = (r, z, a, g, y^*),$$

where r is the set of physically relevant variables (those entering the governing law g); z is the designated cue or non-governing variable; a is the set of stated physical assumptions; g is the governing physical principle (e.g., Newton’s second law, Coulomb’s law); and $y^* \in \mathcal{Y}$ is the solver-verified correct answer.

The *counterfactual family* for template τ is:

$$\mathcal{F}_\tau = \{x_\tau(r, z_i, a)\}_{i=1}^m, \quad (1)$$

where $z_1, \dots, z_m$ are distinct cue values and each $x_\tau(r, z_i, a)$ is a natural-language rendering of the problem. A symbolic or executable verifier certifies the *cue-invariance condition*:

$$y_\tau^*(r, z_i, a) = y_\tau^*(r, z_j, a) \quad \forall i \neq j. \quad (2)$$

3.3 Behavioural Sensitivity Measures

Three complementary sensitivity measures are defined in increasing order of tractability.

Distribution-level sensitivity. Where answer distributions can be estimated by sampling or log-probability approximation over a normalised answer space, the primary distributional sensitivity score is:

$$S_\theta(\tau) = \frac{2}{m(m-1)} \sum_{i < j} D_{\text{JS}}\left(p_\theta(y \mid x_\tau(r, z_i, a)), p_\theta(y \mid x_\tau(r, z_j, a))\right), \quad (3)$$

where $D_{\text{JS}}(\cdot, \cdot)$ is the Jensen–Shannon divergence. Section 11 specifies tractable approximations for open-ended generation.

Answer-flip sensitivity. A robust discrete measure counts the fraction of invariant pairs in which the model’s most likely answer changes:

$$\hat{S}_\theta(\tau) = \frac{1}{\binom{m}{2}} \sum_{i < j} \mathbf{1}[\hat{y}_\theta(x_\tau(r, z_i, a)) \neq \hat{y}_\theta(x_\tau(r, z_j, a))]. \quad (4)$$

Log-probability drop. A third measure captures the maximum degradation in the model’s confidence in the solver-verified answer:

$$S_\theta^{\text{lp}}(\tau) = \max_{j \neq \text{base}} \left[\log p_\theta(y^* \mid x_\tau(r, z_{\text{base}}, a)) - \log p_\theta(y^* \mid x_\tau(r, z_j, a)) \right]. \quad (5)$$

The probe target is the *model-specific sensitivity* $S_\theta(\tau)$, $\hat{S}_\theta(\tau)$, or $S_\theta^{\text{lp}}(\tau)$ —not the prompt-level indicator of which cue value z_i is present.

3.4 Internal Monitor

An internal monitor is a function $f_\phi(h_\theta(x)) \rightarrow \hat{s}$ that maps a hidden-state representation extracted from a prompt $x \in \mathcal{F}_\tau$ to a predicted sensitivity score. The monitor is trained exclusively on training-family activations and evaluated on held-out families. Its prediction target is the model-specific sensitivity of the *family* $\mathcal{F}_\tau$ the prompt belongs to—not the cue value z_i of the particular prompt.

4 Research Questions and Hypotheses

4.1 Primary Research Question

RQ1. Can prompt-side hidden states predict which model–template–family pairs will exhibit counterfactual shortcut sensitivity—measured by output instability across solver-verified invariant families—before any failure is visible in the generated reasoning chain or final answer?

4.2 Complementary Causal Research Question

RQ2. Do localised prompt-side representations contribute causally to dependence on designated physically irrelevant or non-governing cues, as evidenced by necessity and sufficiency interventions on held-out families?

RQ1 is the *reliable core* of the project. RQ2 is a higher-risk objective whose credibility depends on passing all mandatory causal controls (Section 10); a negative or inconclusive answer to RQ2 does not invalidate the project.

4.3 Falsifiable Hypotheses

H1 (Behavioural counterfactual sensitivity). Open-weight language models will exhibit non-trivial output instability across solver-verified physics families in which designated cue edits should not change the correct answer. A *successful* H1 would show that at least one primary model exhibits $\hat{S}_\theta(\tau) > 0$ on a meaningful fraction of template families (indicatively $\geq 20\%$). A *failed* H1 would itself constitute an informative result about that model class’s invariance properties.

H2 (Pre-generation hidden-state predictability). Prompt-side hidden states will predict which model–family pairs will exhibit counterfactual sensitivity better than: (a) prompt-only surface classifiers (TF-IDF, n -gram, statistics-based), (b) cue-span-only classifiers, (c) generic correctness probes, (d) answer confidence and entropy measures, and (e) textual CoT monitors. Indicative go-threshold: AUROC ≥ 0.75 on held-out template families with ≥ 0.10 AUROC advantage over the strongest surface baseline, surviving artefact controls. These are *targets, not promised outcomes*.

H3 (Generalisation beyond artefacts). The hidden-state advantage will survive held-out template families, paraphrase variants, variable renaming, matched length and token statistics, and at least one held-out cue family or physics domain.

H4 (Limited causal contribution). For selected layers and representations, controlled necessity and sufficiency interventions will modulate cue sensitivity on held-out families beyond norm-matched random directions, random sites, nearby-layer controls, and unrelated-family controls. **H4 is explicitly higher risk.** If it fails, the project falls back to a benchmark-plus-monitorability contribution rather than overstating causal conclusions.

4.4 Terminology Conventions

The following terms are used with precise, consistent meanings:

Cue presence. Whether a designated non-governing variable z appears in the prompt. A Level-1 stimulus property.

Shortcut sensitivity / counterfactual instability. The extent to which a specific model’s output changes across invariant family members. A Level-2, model-specific property derived from behaviour.

Shortcut reliance. Reserved for cases supported by *both* behavioural counterfactual evidence (Level 2) *and* appropriately controlled intervention evidence (Level 3). Not used to describe prompt-condition effects alone.

Invariance violation. Family-level sensitivity $\hat{S}_\theta(\tau) > 0$ or equivalent. A descriptive term for the behavioural phenomenon.

We explicitly avoid language claiming that two prompts have “the same answer but different hidden reasoning paths,” because the internal reasoning path is not directly observable. Instead we discuss equivalent-task families, expected physical invariance, observed output instability, and intervention-sensitive internal representations.

5 Benchmark and Counterfactual-Family Construction

5.1 Scope and Physics Domains

The core benchmark will consist of **120–180 canonical counterfactual families**, yielding approximately 900–1,400 rendered prompt instances after controlled numerical instantiations, paraphrases, and variable-renamed variants.

Two core physics domains:

1. **Classical mechanics.** Kinematics, Newton’s laws, work–energy theorem, linear and rotational momentum, circular motion. Strong symbolic templates are available; cue irrelevance can be certified algebraically.
2. **Electrostatics / basic circuits.** Coulomb’s law, electric field, potential, capacitance, Ohm’s law, series/parallel resistance. Similarly clean invariances under non-governing distractor insertion.

Thermodynamics is excluded from the first version. Verbal assumptions, state-variable dependencies, and path-dependent quantities make invariance harder to certify cleanly; it is listed as a future extension (Section 18).

The benchmark uses **text-only, compact problems** with canonical numerical or symbolic answers. Diagrams and multimodal inputs are excluded from the core study.

5.2 Core Cue Families

Priority is given to cue types where irrelevance or non-governance can be formally certified:

1. **Physically irrelevant variable.** A quantity mentioned in the problem that does not enter the governing law for the quantity being asked about. *Example:* the mass of a frictionless, massless pulley in a problem asking for the acceleration of connected masses where only the net force and total mass govern the answer. Irrelevance is certified by symbolic derivation showing the variable does not appear in the derived expression for y^* .
2. **Unit-compatible non-governing distractor.** A quantity with the same units as a governing variable but which plays no causal role in the answer. Certified by showing it does not appear in the derived expression for y^* after all substitutions.
3. **Representation or reference-frame rendering (where the transformation can be verified exactly).** Problems stated in Cartesian vs. polar coordinates, or in different inertial frames, where the correct numerical answer is provably invariant under the transformation. Included only for cases where the equivalence is algebraically exact.

Formula leakage is demoted to Appendix A as a sanity check only. If the problem statement explicitly contains the target formula, the task reduces to lexical retrieval; this is too trivial for a core contribution.

Misleading answer-correlated variables are removed unless the correlation and causal irrelevance can be generated and verified formally without making the task artificial.

Sign and boundary-condition traps are deferred unless their expected invariance or dependence is completely specified and solver-verified.

The cue types are treated as controlled experimental interventions. *Shortcut sensitivity is not assumed from cue presence; it is inferred from model behaviour across the family.*

5.3 Symbolic-Template-First Generation Protocol

All problem families follow a template-first pipeline. An LLM must not invent the underlying physics or determine ground truth.

1. Define a canonical symbolic physics problem: specify relevant variables r , assumptions a , governing principle g , and the solver-derived answer y^* .
2. Identify the designated cue z and provide a symbolic proof sketch or executable certificate that y^* is invariant to z (Eq. 2).
3. Instantiate multiple numerical settings for r , varying z across at least two values per family.
4. Render matched natural-language prompts, ensuring only z changes across family members while r , a , and y^* remain constant.
5. Generate paraphrased and variable-renamed variants for robustness testing.
6. Run automatic checks: dimensional consistency, answer invariance via the parser, assumption completeness, surface-statistic matching (length, number count, variable count, unit count).
7. Run adversarial artefact analysis: prompt a separate model to predict the sensitivity class from the prompt alone; inspect the features it uses.
8. Send every development and test family to dual human validation (Section 6).

Permitted LLM uses: linguistic paraphrasing, variable-name diversification, adversarial artefact discovery, proposing candidate failure cases for human review, and improving naturalness without altering semantics.

Prohibited LLM uses: inventing physics, determining ground truth, defining the correct answer, generating problem semantics that are subsequently adopted without verification.

5.4 Answer Parsing and Normalisation

A rule-based answer parser will extract the numerical or symbolic answer from model output. The parser must: (a) handle unit variations and equivalent algebraic forms; (b) apply the same normalisation to all members of a family; and (c) be validated on a sample of human-verified answers before any sensitivity computation. All sensitivity measures use the normalised answer, not raw output text.

6 Validation and Artefact-Control Protocol

6.1 Human Validation

Every development and test family will undergo **dual independent validation** by reviewers with appropriate physics competence (undergraduate physics background minimum for the relevant domain). Validators will assess:

1. Is the designated cue truly irrelevant or non-governing for this problem?
2. Are the stated assumptions sufficient to uniquely determine the answer?
3. Is the correct answer invariant across all family members as claimed?
4. Does each family member change only the intended variable, with no unintended co-variation?
5. Is the wording natural and free from inadvertent difficulty shifts or additional cues?

Disagreements are resolved by a third adjudicator. Agreement on categorical judgements will be measured by Cohen’s κ , targeting $\kappa \geq 0.80$ on the development set. This is a target; any shortfall will be reported and analysed. Solver derivation metadata will be stored with every family.

6.2 Mandatory Artefact Controls

The following controls must be reported before any hidden-state result is interpreted as meaningful. If any surface control matches or exceeds the hidden-state probe on held-out families, the result is not yet deconfounded.

Surface classifiers. TF-IDF logistic classifier; bag-of-words / n -gram classifier; cue-span-only classifier.

Statistics baselines. Prompt length; equation count; variable count; number count; unit token count; stylometric features.

Matched controls. Length-matched prompts; formula-count-matched prompts; variable-count-matched prompts; unit-token-matched prompts; style- and paraphrase-matched controls.

Semantic controls. Semantically irrelevant distractor controls (a variable added to the prompt that does not constitute the designated cue); same-answer/no-cue controls (families with matching answer but no cue manipulation); variable-renaming controls.

Answer-range controls. Answer-range balancing across train/test; no-cue difficulty-matched families.

Embedding baseline. Embedding-layer or earliest-layer hidden-state baseline: if this matches the target layer, the signal is likely lexical rather than representational.

Where appropriate, we will residualise or condition on measured surface covariates and test whether the hidden-state advantage survives.

6.3 Data Splitting

Splits are by canonical template family, never by rendered prompt instance. No paraphrase, numeric sibling, or variable-renamed version of a test family may appear in training or development data.

Evaluation will include:

- Held-out template-family evaluation (primary generalisation test).
- Held-out paraphrase evaluation.
- Held-out numerical instantiation evaluation.
- Leave-one-cue-family-out evaluation (where feasible given family count).
- Leave-one-domain-out evaluation as a robustness test.
- Cross-domain transfer where scientifically meaningful.

7 Model Selection and Computational Resources

7.1 Available Computational Resources

All experiments will use the **Sharanga cluster** with the following access patterns:

- Up to 2 H200 GPUs at a time.
- Up to 3 H100 GPUs at a time.
- Up to 2 A100 GPUs at a time.

No commercial API credits are required or assumed. Access to closed-weight models may be added as an optional external-validity supplement if it later becomes available, but it is not a dependency. The open-weight strategy is presented as a *reproducibility strength*: every result will be reproducible by any researcher with equivalent open GPU resources.

7.2 Model Roles

Models serve one of three distinct roles that must not be conflated:

Full hidden-state and causal analysis (primary). Two open-weight dense models from different architectural families in the 7–14B parameter range, selected for mature activation-hooking support, stable hidden-state extraction, compatibility with intervention tooling, and strong instruction-following. A candidate structure is one Qwen-family dense model and one Llama- or Mistral-family dense model. Final selection will be made at the start of Stage 2 (Section 12) based on current availability, tokeniser and architecture support, reproducibility, and licensing.

Behavioural replication (secondary). One reasoning-tuned open-weight model in the 7–32B range, used to test whether the shortcut sensitivity phenomenon extends to models with explicit reasoning optimisation. Activation access is required for this model; it may be added to the full causal analysis if tooling is proven stable.

Behavioural external validity and judge baselines (large model). A Qwen3.5-397B-class model or similar very large locally hosted open model, used *primarily* for: (a) behavioural counterfactual evaluation at scale; (b) open-weight textual-monitor and

judge baselines; (c) adversarial artefact search; (d) controlled paraphrasing; (e) assessing whether sensitivity persists in a substantially stronger model. This model is **not** the primary activation-patching target because MoE routing and activation scale introduce major interpretability complications. Activation patching on a model of this class is excluded unless its tooling is separately validated.

Table 2: Model roles in the PhysMon study. Final model versions are not hard-coded; selection criteria are given.

Role	Candidate family / range	Primary use in this project
Dense model A (internals)	Qwen-family, 7–14B	Activation extraction, probing, causal intervention
Dense model B (internals)	Llama- or Mistral-family, 7–14B	Same as A; cross-architecture replication
Reasoning-tuned model	Reasoning-distilled open model, 7–32B	External validity; cross-model sensitivity replication; activation analysis if tooling is verified stable
Large open model	Qwen3.5-397B-class or equivalent	Behavioural evaluation; judge baselines; adversarial artefact search; paraphrasing

7.3 Activation Extraction Tooling

Activation extraction will use TransformerLens or an equivalent stable hooking library. Before the main study begins, we will: (a) confirm hook support on both primary models; (b) validate log-probability extraction and deterministic generation settings; (c) reproduce a known probing result (e.g., from a standard hidden-state verification benchmark) as a calibration check; and (d) estimate activation storage and compute costs per model and per stage.

8 Hidden-State Monitoring Methodology

8.1 Staged Activation Extraction

Rather than extracting activations at every layer and token position uniformly, we follow a **staged, hypothesis-driven hierarchy**:

1. **Primary sites (prompt-side, always extracted).** Residual stream at the designated cue-token position or pooled cue span; residual stream at the final prompt token before generation begins; attention-output and MLP-output representations at those sites.
2. **Secondary sites (optional, post-pilot).** Early generated-token states, only after the prompt-side result is established and only using teacher-forced, semantically aligned positions or a similarly defensible alignment scheme. Independently sampled

CoT traces are poorly aligned across runs and are therefore excluded from the default intervention plan.

The intended strongest result is: *prompt-side activations predict later counterfactual instability before chain-of-thought or answer generation begins*. This is more novel than a post-hoc monitor over already-generated reasoning. Activation pooling strategy, normalisation, storage precision, batching, and model-specific layer mapping will all be documented for reproducibility.

8.2 Probe Architecture and Training

We will use **simple, interpretability-resistant probe architectures first**:

- L_2 -regularised logistic regression for binary sensitivity targets.
- Linear regression for continuous sensitivity targets ($\hat{S}_\theta(\tau)$ or $S_\theta^{\text{lp}}(\tau)$).
- Calibrated linear probes with Platt scaling.
- An optional non-parametric hidden-state verifier baseline for comparison.

Training procedure:

- Nested hyperparameter selection using training families only (no test family leakage at any stage).
- Family-level cross-validation (split unit: canonical template family, not rendered instance).
- Feature standardisation fitted on training data only.
- Calibration on development families.
- Class-imbalance handling via re-weighting where necessary.

8.3 Probe Targets

Primary target. Family-level counterfactual shortcut sensitivity $\hat{S}_\theta(\tau)$ or continuous equivalent on held-out families.

Secondary targets. Answer flip under invariant edits; change in probability assigned to the invariant correct answer; output-distribution divergence under cue edit; family-level invariance violation; cue-dependent correctness.

Auxiliary diagnostic (not the central claim). Prompt condition (which cue value is present) may be included as an auxiliary probe to measure how much prompt-condition information overlaps with sensitivity information—specifically, to demonstrate that the two are not identical.

Generic correctness probe baseline. A probe predicting simple answer correctness will be included to confirm that the shortcut-sensitivity monitor is not merely detecting generic problem difficulty. H2 requires the sensitivity probe to outperform both surface baselines *and* the generic correctness probe on held-out families.

9 Baselines

A reviewer-proof baseline suite is mandatory. If any baseline performs comparably to the hidden-state probe after artefact controls, the result must be reframed accordingly before submission. Table 3 lists all required baselines.

Table 3: Required baselines and artefact controls. All rows marked “Yes” must appear in the main paper.

Baseline / Control	Purpose	Mandatory?
<i>Surface and prompt baselines</i>		
TF-IDF logistic classifier	Labels readable from surface text alone?	Yes
Bag-of-words / n -gram classifier	Structural surface features	Yes
Cue-span-only classifier	Cue text itself reveals sensitivity?	Yes
Prompt-statistics classifier (length, number count, variable count, equation count, unit-token count)	Formatting confounds	Yes
Embedding-layer / earliest-layer hidden-state	Lexical signal from first layer?	Yes
<i>Behavioural baselines</i>		
Answer confidence (log-probability)	Standard confidence proxy	Yes
Entropy / uncertainty over answer tokens	Standard weak baseline	Yes
Self-consistency (majority vote over samples)	Stability indicator	Yes
Black-box counterfactual sensitivity (two-query perturbation)	Does a cheap oracle already solve the task?	Yes
Generic correctness probe	Distinguishes sensitivity from generic difficulty	Yes
<i>Textual-monitor baselines</i>		
CoT text classifier	How much does text alone reveal?	Yes
Answer-and-rationale classifier	Full output vs. activations	Yes
Open-weight LLM monitor (≥ 2 judge models or settings)	Judge instability must be measured	Yes
Critic / process-reward-model-style baseline	Standard reasoning verifier	Recommended
<i>Causal intervention controls</i>		

Table 3 continued.

Baseline / Control	Purpose	Mandatory?
Norm-matched random direction (patching)	Baseline for any intervention claim	Yes
Random patch site	Localisation control	Yes
Nearby-layer control	Specificity to identified site	Yes
Unrelated-family donor	Construct specificity	Yes
Same-answer/no-cue family donor	Answer-bias control	Yes
<i>Optional additional baselines</i>		
Non-parametric / representation-verification baseline	Modern hidden-state comparison	Optional
SAE feature-based baseline	After core result established; high artefact risk	Optional
Attention / saliency on cue tokens	Inexpensive attribution comparator	Optional

Judge-baseline handling. At least two distinct judge models or prompt settings will be used for LLM-as-judge baselines, and judge sensitivity (performance range across choices) will be reported explicitly. No single judge will be treated as ground truth. Because no commercial APIs are assumed, locally hosted open-weight models will serve as judge baselines.

10 Causal Intervention Methodology

10.1 Scope of the Causal Claim

Activation patching alone does not prove that a representation constitutes the model’s “shortcut reasoning path.” The causal objective is limited and precisely stated:

There exist localised prompt-side representations whose intervention changes the model’s sensitivity to designated irrelevant-cue edits on held-out, solver-verified physics problem families, indicating that these representations contribute causally to cue dependence.

This claim does not extend to: identification of a complete shortcut circuit; recovery of the model’s true reasoning path; or proof that the model “used the shortcut” in any philosophically strong sense. We explicitly avoid path language throughout.

10.2 Necessity Tests

At probe-identified prompt-side sites, we will test:

1. **Direction projection / concept erasure.** Project out the probe-identified direction from the residual stream. Measure whether cue sensitivity decreases while general answer capability is preserved.
2. **Activation ablation.** Zero out or mean-ablate activations at selected sites. Report effects on sensitivity and on general accuracy.
3. **Targeted resampling / patching.** Resample activations from unrelated families and measure the effect on sensitivity.

10.3 Sufficiency Tests

1. **Direction injection.** Inject a calibrated version of a high-sensitivity direction into low-sensitivity examples; measure whether cue dependence increases in the predicted direction.
2. **Cross-family patching.** Patch activations from high-sensitivity families into matched low-sensitivity runs at probe-identified sites; test whether sensitivity increases.

10.4 Mandatory Controls

Every causal experiment must include all of the following:

1. Norm-matched random direction (same ℓ_2 norm, random orientation).
2. Random patch site (randomly chosen layer, same token position).
3. Nearby-layer control (adjacent layer, same position).
4. Unrelated-family donor (activations from a family in the other physics domain).
5. Same-answer/no-cue family donor (activations from a family where the correct answer matches but no cue manipulation is present).
6. Cue-presence-matched but behaviourally stable donor (activations from a family that contains the cue value but shows low sensitivity).
7. General-performance damage control (verify overall answer accuracy on unrelated families is not degraded).
8. Answer-bias control (verify the intervention does not shift the model toward one answer regardless of physics).
9. Multiple intervention strengths (characterise the dose-response curve).
10. Held-out template-family evaluation (effects measured on families not used to identify the intervention site).

10.5 Fallback Protocol

If intervention effects are weak, unstable, or fail necessity/sufficiency controls, the project will **explicitly pivot** to a diagnostic monitorability claim. Recent work shows that strong latent signals can be diagnostic while resisting reliable control (et al., 2024); this outcome is itself a scientifically informative result that will be reported as such, not suppressed.

11 Evaluation Metrics and Statistical Analysis

11.1 Primary Monitoring Metrics

- **AUROC** over held-out template families (primary monitoring metric).
- **AUPRC** (critical when sensitivity labels are imbalanced).
- **Expected calibration error** and reliability diagrams.
- **Brier score**.
- **FNR at a pre-registered fixed FPR**: the operating point is fixed before evaluating on test families.

False-negative rate comparisons are reported as *derived analyses* at the pre-registered operating point, not as the sole or primary headline metric.

11.2 Primary Behavioural Metrics

- Family-level answer-flip rate $\hat{S}_\theta(\tau)$ (Eq. 4).
- Change in normalised correct-answer log-probability $S_\theta^{\text{lp}}(\tau)$ (Eq. 5).
- Jensen–Shannon divergence between answer distributions where tractable (Eq. 3).
- Family-level counterfactual degradation: fraction of families where model accuracy drops under cue edits.

On distribution estimation. Full answer-distribution estimation via sampling is expensive. Tractable approximations will use: (a) the model’s top- k token probabilities over a normalised answer vocabulary, and/or (b) repeated sampling with temperature to estimate the answer distribution. The chosen approximation will be documented and validated on a held-out subset before deployment.

11.3 Causal Metrics

- Change in $\hat{S}_\theta(\tau)$ under necessity and sufficiency interventions (reported separately).
- Effect size relative to all mandatory controls.
- Preservation of general task accuracy (per-family accuracy on unrelated families).
- Specificity to held-out cue families (intervention must transfer, not merely fit training families).

11.4 Statistical Analysis

Inference unit. The *canonical template family* is the independent unit for all statistical inference. Aggregating over rendered instances without accounting for family-level clustering inflates effective sample size and will not be done.

Confidence intervals. Bootstrap confidence intervals over canonical families for all primary metrics.

Baseline comparisons. Paired randomisation tests or Wilcoxon signed-rank tests over matched canonical families.

Layer-wise and multi-probe comparisons. Benjamini–Hochberg false-discovery-rate correction for layer-wise probe scans and multiple hypothesis comparisons.

Effect sizes. Reported alongside p -values for all primary claims; a statistically significant but negligible effect size is not a positive result.

Probe layer selection. A pre-specified primary layer set (e.g., final quarter of the network) or a nested selection procedure using only training and development families will be documented *before* evaluating on test families, to prevent layer cherry-picking.

Seed sensitivity. All key results will be re-run with at least two random seeds; variance across seeds will be reported.

Calibration uncertainty. Calibration error will be reported with bootstrap uncertainty for all monitoring claims.

12 Ordered Research Execution Plan and Decision Gates

The following 11 stages are ordered by scientific dependency. **No dates, durations, or calendar deadlines are assigned.** Each stage carries an explicit decision gate.

Stage 1 — Formalise the Construct. *Tasks:* Define cue irrelevance, non-governance, expected invariance, shortcut sensitivity, and the boundaries of allowable causal claims. Write the construct-validity specification using the notation in Section 3. Specify the exact sensitivity measures, answer parser, and exclusion criteria. Finalise H1–H4.

Outputs: Construct-validity specification document; notation reference used consistently across all later stages.

Decision gate: **Stop or redesign** if shortcut sensitivity cannot be defined objectively and measured without reliance on prompt-condition labels alone.

Stage 2 — Validate Technical Instrumentation. *Tasks:* Confirm stable activation extraction and intervention hooks on both selected dense open models. Reproduce a known probing and patching result (e.g., from a published hidden-state verification benchmark) as a calibration check. Validate answer parsing and log-probability extraction. Validate deterministic generation settings. Estimate activation storage and compute costs.

Outputs: Instrumentation validation report; compute budget estimate.

Decision gate: **Change models or tooling** if activation extraction or intervention support is unreliable on either primary model.

Stage 3 — Construct the Pilot Benchmark. *Tasks:* Build a mechanics-first pilot of approximately 30 canonical families. Include two core cue types (physically irrelevant variable; unit-compatible non-governing distractor). Implement symbolic/executable verification for all pilot families. Conduct dual human review for all pilot families. Construct prompt-only and artefact baselines *before* any hidden-state analysis begins. *Outputs:* Pilot dataset with solver metadata; human-validation reports; artefact-baseline performance sheet.

Decision gate: **Redesign** if: (a) invariance certification fails for more than 10% of pilot families; (b) validator disagreement on cue irrelevance gives $\kappa < 0.65$; or (c) prompt-only baselines trivially recover the target (TF-IDF AUROC ≥ 0.85 on held-out pilot families).

Stage 4 — Establish Behavioural Sensitivity. *Tasks:* Run both primary models across all invariant pilot-family variants. Measure $\hat{S}_\theta(\tau)$ and $S_\theta^{\text{lp}}(\tau)$ per family. Verify that sensitivity is repeatable across seeds. Separate cue sensitivity from generic inaccuracy and difficulty using the generic correctness probe.

Outputs: Pilot sensitivity distribution per model; sensitivity vs. difficulty analysis.

Decision gate:

- **Stop:** fewer than 10% of pilot families show $\hat{S}_\theta > 0$ in either primary model.
- **Proceed cautiously / redesign problem families:** sensitivity rate between 10% and 20%.
- **Proceed:** at least one primary model shows $\hat{S}_\theta > 0$ on $\geq 20\%$ of pilot families.

Stage 5 — Test Hidden-State Monitoring. *Tasks:* Train prompt-side linear probes on family-level behavioural sensitivity using training-family activations. Evaluate on held-out pilot families. Compare against all mandatory surface, correctness, text, confidence, and black-box baselines. Run paraphrase, variable-renaming, and statistics-matching artefact tests.

Outputs: Per-layer AUROC and AUPRC; baseline comparison table; artefact control results.

Decision gate: **Do not expand the benchmark** if the hidden-state monitor matches prompt-only or cue-only baselines after artefact controls. Indicative go-threshold: AUROC ≥ 0.75 on held-out families with ≥ 0.10 AUROC advantage over the strongest surface baseline, surviving paraphrase and renaming controls.

Stage 6 — Expand and Validate the Benchmark. *Tasks:* Expand to 120–180 canonical families. Add the electrostatics/circuits domain. Add the representation/frame cue type if the Stage 3 pilot indicates it is tractable. Finalise family-level development and test splits. Complete dual human validation and solver metadata for all development and test families. **Freeze the benchmark protocol** before the central intervention study begins.

Outputs: Full benchmark with validation reports; final split files; solver proof metadata.

Decision gate: **Do not proceed to mechanistic claims** if artefact controls degrade or if annotation agreement falls below target on the expanded set.

Stage 7 — Run the Full Monitoring Study. *Tasks:* Perform layer- and site-specific monitoring on the full benchmark. Evaluate calibration and cross-family transfer. Test held-out cue families and held-out domain. Compare the two primary dense models and the reasoning-tuned replication model. Run behavioural-only comparison using the large open model.

Outputs: Full monitoring results table; calibration curves; cross-model comparison; cross-domain transfer results.

Decision gate: **Narrow claims** if the monitor generalises only within a single template type or model family. Proceed to Stage 8 only if the monitor shows robust held-out-family generalisation.

Stage 8 — Conduct Controlled Causal Interventions. *Tasks:* Select intervention sites based only on training/development data. Run necessity experiments (direction projection, ablation, resampling) at probe-identified prompt-side sites. Run sufficiency

experiments (direction injection, cross-family patching). Run all mandatory controls (Section 10). Attempt replication of the strongest effect in the second primary model. *Outputs:* Intervention effect tables with controls; necessity/sufficiency effect sizes; replication results.

Decision gate: **Retain causal language only if** effects are (a) directionally correct, (b) statistically reliable across seeds and models, (c) specific to the designated construct, and (d) clearly stronger than all random-direction and random-site controls on held-out families. Otherwise, pivot to diagnostic framing without causal language.

Stage 9 — Stress-Test Robustness and Alternative Explanations. *Tasks:* Residualise surface features and re-test hidden-state advantage. Test answer-bias, generic-correctness, and generic-difficulty explanations. Run leave-one-cue-family-out and leave-one-domain-out analyses. Inspect failure cases. Conduct human audit of a sample of ≥ 30 monitor successes and failures.

Outputs: Robustness tables; failure-case taxonomy; human audit report.

Stage 10 — Freeze Paper Claim and Select Venue Framing. If causal evidence is strong:

Lead with localised representations contributing to cue sensitivity. Consider ICLR as a stretch target. Retain benchmark and monitoring as major co-contributions.

If causal evidence is weak but monitoring is strong: Frame as counterfactual scientific-reasoning monitorability. Target ACL, EMNLP, or NAACL. Report diagnostic-but-not-controllable representations as a scientifically informative result.

If hidden-state advantage is weak: Pivot to a benchmark and behavioural study only if the benchmark reveals a strong, novel CoT-faithfulness or counterfactual-sensitivity finding. Otherwise, stop rather than forcing a weak paper.

Stage 11 — Reproducibility and Release Preparation. *Tasks:* Freeze all configurations and random seeds. Prepare a benchmark card. Package and test all code (family generation, evaluation, probing, intervention). Audit licenses. Prepare human-validation documentation. Prepare an internal paper draft.

Outputs: Code repository with full reproducibility; benchmark card; internal paper draft.

Decision gate: **Do not release the benchmark publicly** before the core hidden-state analysis paper is internally complete. Releasing the benchmark alone risks enabling other groups to attach a competing analysis to the data.

13 Go/No-Go Criteria

Table 4 summarises the project-level go/no-go criteria.

Table 4: Go/no-go criteria for the full PhysMon study.

Criterion	Evidence to proceed	Evidence requiring redesign	Evidence requiring stop / pivot
1. Behavioural sensitivity exists	≥ 1 model shows $\hat{S}_\theta > 0$ on $\geq 20\%$ of pilot families	Sensitivity 10–20%; redesign problem families before expansion	$< 10\%$ of families show any sensitivity in either model
2. Dataset invariance holds	Symbolic verifier passes; $\kappa \geq 0.80$ on categorical validation items	Disagreement in 15–25% of families; revise affected templates	$> 25\%$ of families fail invariance check or annotation
3. Prompt-only artefacts do not explain the result	TF-IDF / statistics AUROC < 0.65 on held-out families; hidden-state advantage ≥ 0.10 AUROC after matching controls	Artefact baselines match hidden states on some cue types; add more controls	Artefact baselines match or exceed hidden states across all conditions
4. Hidden-state advantage survives held-out families	AUROC ≥ 0.75 on held-out template families in ≥ 1 primary model	AUROC in 0.65–0.75 range; marginal advantage; strengthen controls	AUROC at or below best surface baseline
5. Intervention exceeds random controls	Necessity and sufficiency effects directionally correct and reliably stronger than all random-direction and random-site controls	Effects present but statistically unreliable; widen to more families	No reliable gap over any random control
6. Cross-model / cross-family replication	Effect replicates in both primary models and on held-out cue families	Replication in one model only; narrow the causal claim	Effect is model-idiosyncratic and does not replicate

Table 4 continued from previous page.

Criterion	Evidence to proceed	Evidence requiring redesign	Evidence requiring stop / pivot
7. Not reducible to generic difficulty	Generic correctness probe significantly weaker than shortcut-sensitivity probe on held-out families	Generic probe explains 50–75% of the variance; add residualisation and re-test	Generic correctness probe fully explains the hidden-state advantage

14 Risks, Fallback Paths, and Scope Boundaries

14.1 Construct-Validity Risk

Risk: The monitor learns to predict prompt condition, problem formatting, or generic difficulty rather than model-specific shortcut sensitivity. *Mitigation:* Behaviourally derived target labels; matched artefact controls; residualisation of surface covariates; family-level held-out splits; mandatory prompt-only baseline suite (Section 9). If the monitor advantage collapses after artefact controls, the claim is narrowed to match the surviving evidence.

14.2 Behavioural-Signal Risk

Risk: Strong models may exhibit too little cue sensitivity to train a meaningful monitor. *Mitigation:* Explicit pilot stop criteria (criterion 1 in Table 4); multiple model scales; a range of cue types targeting sensitivity at different model strengths. The project will not manufacture sensitivity through unnatural prompts; genuine invariance in a model class is an informative result.

14.3 Causal-Control Risk

Risk: Diagnostic representations may not be controllable; interventions may fail necessity or sufficiency controls. *Mitigation:* Explicit fallback to diagnostic monitorability framing (Stage 10 decision gate). A strong benchmark-plus-monitor paper remains publishable without causal evidence.

14.4 Dataset-Validity Risk

Risk: Subtle physics errors or implicit assumption drift may invalidate families despite symbolic generation. *Mitigation:* Symbolic-template-first generation; solver proof meta-data; dual human validation for all test families; third-adjudicator protocol; adversarial artefact search.

14.5 Scaling Risk

Risk: Very large MoE models may be unsuitable for activation patching due to routing complexity and storage scale. *Mitigation:* Large models are restricted to behavioural and judge roles by default. Dense 7–14B models are the activation-analysis targets.

14.6 Novelty and Scoop Risk

Risk: Adjacent work on hidden-state monitoring and shortcut benchmarks is progressing rapidly in 2025–2026. *Mitigation:* The project centres on *solver-verified physics counterfactual sensitivity*—a specific combination not yet published. The benchmark and hidden-state result will be developed together and released simultaneously. An internal paper draft will be prepared before any public benchmark release. Scope will be kept deliberately narrow.

14.7 Explicit Scope Boundaries

The following are **out of scope** for the core paper:

- Diagrams, multimodal inputs, or visual physics problems.
- Thermodynamics in v1.
- Agentic or multi-agent scientific reasoning applications.
- Solver-escalation or tool-use routing systems.
- Model diffing or fine-tuning comparative studies.
- Claims about the model’s “true reasoning path” or “genuine first-principles reasoning.”

15 Expected Contributions

All contributions are stated contingently on empirical success at the relevant decision gate.

1. **Construct contribution (unconditional).** A precise behavioural definition of model-specific physics shortcut sensitivity based on solver-verified counterfactual invariance, with formal notation and a clear three-level distinction between prompt condition, observed model behaviour, and internal representation.
2. **Benchmark contribution (contingent on dataset validity).** A family-structured, solver-verified benchmark of 120–180 canonical physics counterfactual families with proof metadata, dual-validator annotation, matched artefact controls, and template-level splits. Unlike existing physics benchmarks that target only aggregate accuracy, this benchmark is designed for internal analysis of sensitivity.
3. **Monitoring contribution (contingent on H2 and H3).** Evidence that prompt-side hidden states predict future counterfactual sensitivity beyond surface, textual, confidence, and black-box baselines on held-out template families, constituting a form of *pre-generation scientific-reasoning monitorability*.
4. **Causal representation contribution (contingent on H4 and all mandatory controls).** Limited evidence that selected prompt-side representations contribute

causally to cue dependence, supported by necessity, sufficiency, and all mandatory random controls. This claim will be narrowed or retracted if controls fail.

5. **Methodological contribution (unconditional).** A rigorous evaluation protocol for separating prompt-condition decoding from behaviourally grounded internal monitoring, including: artefact control requirements, family-level statistical analysis, and go/no-go criteria applicable beyond the physics domain.
6. **Scientific-reasoning contribution (contingent on behavioural study quality).** A demonstration that physics—as a verifier-rich substrate—enables systematic study of *silent brittleness*: cases where models produce correct, well-reasoned outputs but harbour hidden sensitivity to physically irrelevant cues that becomes apparent under controlled counterfactual edits.

Neither the benchmark alone nor generic hidden-state superiority will be claimed as the main novelty. The claim is their *combination* under solver-verified physics invariances with rigorous artefact controls.

16 Publication and Venue Strategy

Table 5: Venue strategy. ACL/EMNLP/NAACL are the primary realistic targets; ICLR is a stretch target requiring substantially stronger causal evidence.

Venue	Primary framing	Likely reviewer objections	Must-have evidence
ACL / EMNLP / NAACL	CoT faithfulness; scientific-reasoning evaluation; hidden-state monitorability; behaviourally grounded counterfactual sensitivity	Construct validity; dataset artefacts; judge-baseline stability; generalisation breadth	Benchmark quality; held-out family generalisation; strong prompt-only and text baselines; clean scientific-reasoning story
ICLR	Internal representations; deconfounded representation analysis; necessity and sufficiency; causal contribution to cue dependence	Probe-only result; weak localisation; patching overclaim; narrow domain; no cross-model replication	Robust intervention effect; negative controls; cross-model replication; no prompt-artefact explanation; restrained mechanistic claims
NeurIPS / ICML	Scalable monitorability; broader reasoning generalisation	Narrow physics framing; insufficient generality; limited methodological novelty	Broader domain coverage or a scalable monitoring result; more model classes

Primary targets: ACL / EMNLP / NAACL. If monitoring is strong and causal evidence is modest or absent, the paper will be framed around counterfactual scientific-reasoning monitorability: physics as a verifier-rich substrate for detecting silent brittleness before visible failure. This framing aligns with CoT-faithfulness and scientific-reasoning evaluation tracks at these venues.

Stretch target: ICLR. ICLR requires unusually strong, deconfounded, cross-model causal evidence. This target will be pursued only if: intervention effects are robust across both primary models; necessity and sufficiency controls pass; and cross-family replication is demonstrated.

Minimum publishable unit. The minimum publishable unit is: a solver-verified physics counterfactual benchmark; a behavioural study showing model-specific cue sensitivity and CoT faithfulness gaps; and strong prompt-only and judge baselines. This supports a respectable ACL/EMNLP submission even without causal evidence.

17 Reproducibility and Release Plan

- All configurations, hyperparameters, and random seeds will be frozen before test evaluation and documented in an accompanying configuration file.
- A **benchmark card** will document dataset scope, construction pipeline, physics domains, cue types, invariance certification procedure, human-validation protocol, known limitations, and intended uses.
- Code will be released for: canonical family generation, symbolic verification, prompt rendering, answer parsing, evaluation metrics, probe training, and causal interventions.
- All solver derivations and proof metadata will be released alongside the benchmark data.
- Human-validation documentation will be prepared consistent with responsible data release guidelines.
- Licenses for all model weights, datasets, and software dependencies will be audited before release.
- **The benchmark will not be released publicly before the core hidden-state analysis paper is internally complete**, to prevent competing groups from attaching their own analysis to the data before the originating work is published.

18 Optional Future Extensions

The following extensions are **explicitly deferred** from the core project. They are listed here only to establish scope boundaries and must not influence the core design, resource plan, or go/no-go criteria.

Scale-up. Running the same probing and patching pipeline on 70B-scale open models. Straightforward once the primary pipeline is complete.

Thermodynamics and additional domains. Expanding the benchmark after the core two-domain study is validated.

Multimodal physics. Extending to diagram-based or multimodal problem variants after the text-only construct is validated.

Model diffing and fine-tuning comparison. Comparing base vs. physics-fine-tuned models using the same hidden-state infrastructure. This would form a distinct second paper.

Runtime monitoring application. Using the hidden-state monitor as a trigger for escalating to a symbolic physics solver or verifier-agent in a scientific reasoning pipeline. This is a future system application, not a core mechanistic contribution.

Cross-domain transfer. Testing whether a monitor trained on physics sensitivity generalises to chemistry, mathematics, or engineering problem families with analogous verifiable structure.

A Formula Leakage as a Sanity Check

Formula leakage—the condition in which the target formula is stated explicitly in the problem setup—was a core shortcut type in an earlier version of this proposal. It is demoted to an appendix sanity check for the following reason.

If a problem explicitly contains the target formula, the task reduces to lexical retrieval rather than shortcut reasoning in any scientifically interesting sense. Any model that matches the formula in the problem to the formula needed for the answer is engaging in a plausible and arguably correct strategy, not exploiting an irrelevant cue. This means formula-leakage sensitivity does not constitute evidence of shortcut reliance in the sense defined in Section 3.

We will include 5–10 formula-leakage family pairs in an appendix to confirm that the answer parser and probing infrastructure function correctly on a trivially detectable condition. If activation probes detect formula-leakage sensitivity substantially below chance, this would indicate a problem with the probing setup. If they detect it well above chance, this is expected and provides no meaningful contribution. Results from this sanity check will be reported in a supplementary table only.

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